

UG (Sem-I) Examination-2025

(NEP : 2020)

AC-I : PHYSICS

Time : 3 hours

Full Marks : 60

Pass Marks : 24

*Candidates are required to give their answers in
their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer from both the Groups as directed.

Group—A

(Compulsory)

Very Short Answer Type Questions

1. Answer the following very short answer type
questions : $1 \times 5 = 5$

(a) What is a vector quantity?

S343/79/36

(1)

(Turn over)

(b) Write second order homogeneous differential equation with constant coefficient.

(c) What is Hooke's law?

(d) What is angle of contact?

(e) What is surface energy?

Short Answer Type Questions

2. Differentiate between Young's modulus of elasticity and Bulk modulus of elasticity. 5
3. State and prove law of conservation of energy. 5

Group - B

(Long Answer Type Questions)

Answer any three questions of the following : 15×3=45

4. (a) Describe damped and forced oscillations.
- (b) Solve the differential equation

$$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$$

S343/79/36

(2)

Contd.

(c) Define surface tension. What are the effects of temperature and pressure on surface tension?

5. Establish relationship among different elastic constants.
6. Derive Poiseuille's equation for the flow of a liquid through a capillary tube. How it was corrected?
7. State and explain Kepler's laws of planetary motion.
8. (a) Describe centrifugal and Coriolis force.
- (b) State and prove law of conservation of energy.
- (c) Derive an expression for velocity in spherical coordinate system.

————— x —————

S343/79/36

(3)

(P-700)

8. (a) Solve the differential equation :

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$$

- (b) Find the angle between the two vectors:

$$\vec{a} = 3\vec{i} + 4\vec{j}$$

$$\vec{b} = 4\vec{i} + 3\vec{j}$$

- (c) Define viscosity. What are the effects of temperature and pressure on the viscosity of a liquid.

————— x —————

UG (Sem-I) Examination-2025

(NEP : 2020) (Ex Student)

MN-I : PHYSICS

Time : 3 hours

Full Marks : 60

Pass Marks : 24

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Groups as directed.

Group—A

(Compulsory)

Very Short Answer Type Questions

1. Answer the following very short answer type questions : 1 × 5 = 5

- (a) What do you mean by conservation of momentum?

- (b) Define a vector quantity.
- (c) Define modulus of rigidity.
- (d) If x be the displacement for a velocity of 5 units, then express it into a first order differential equation.
- (e) Write any one Kepler's law of planetary motion.

Short Answer Type Questions

- 2. State Hooke's law of elasticity. Differentiate between longitudinal and shearing strain with suitable diagram. 5
- 3. What is non-inertial frame of reference? Explain centrifugal and Coriolis force for a rotating frame of reference. 5

Group - B

(Long Answer Type Questions)

S344/99/234

(2)

Contd.

Answer any **three** questions of the following :

15×3=45

- 4. Establish relationship among different elastic constants.
- 5. Define surface tension and surface energy. Derive expression for surface energy. Deduce an expression for excess pressure inside a liquid drop.
- 6. What is damped oscillation? Derive an expression for it. Explain critical damping.
- 7. (a) What is central force?
(b) Explain angular momentum and its conservation.
(c) Explain spherical and cylindrical coordinate system with diagram.

S344/99/234

(3)

(Turn over)